F723 — Move 1 (source e=f): the make-or-break identification, run on the FG/K820 structure. e=f reduces to ONE target-innocent structural identification — "the electroweak-symmetric vacuum = the geometric base point (Jordan cone identity e)" — and that identification forces f=e by a dimension count (both $SU(2)^2$ and Stab(e) are SO(4)⊂SO(5), so EW-fixes-e ? EW ⊆ Stab(e) ? EW = Stab(e) = the Jordan SO(4)). This SEPARATES from BST's deepest open problem (the full SM gauge embedding / the dim-11→12 color question, gauge-note §4): it needs only the vacuum identification, NOT the SU(3) color assignment. So the muon can bank without solving the full embedding — Cal §125's sharp sub-question answered YES, it separates. BUT the identification itself is a target-innocent structural INPUT (muon-mass-free, physically natural: the geometric base point is the maximally-symmetric vacuum), not yet a theorem (proving it needs the Higgs-potential-minimum-sits-at-the-base-point dynamics, genuinely deep). So: e=f is DERIVED-CONDITIONAL on "geometric vacuum = EW vacuum," a single shallow target-innocent condition — a large reduction from K979's "conditional on the deepest open problem." I do NOT fire K967; whether "Derived-conditional on (base point = EW vacuum)" banks the muon Derived or holds it IDENTIFIED is Keeper+Cal's ruling. Framed for them, pinned to source, derived not matched.

Lyra, Wed 2026-07-29 07:41 EDT. Move 1, corpus-first. The two-space hazard resolved (electroweak SU(2) is in the compact SO(5), same space as e), and e=f localized to one identification that separates from the color problem. Cal #27 held: I flag it as an identification, not a closure.

Step 0 — resolve the two-space hazard (which SO(4), which space?)

There are two "SO(4)"s in play, and conflating them is the F722-era risk. Pinned from the corpus: - The electroweak **SU(2)_L×SU(2)_R that labels O** is inside the compact SO(5) (K = SO(5)×SO(2)): the SO(5) vector 5 = (2,2)⊕(1,1) (F603/F633) and the SO(5) spinor 4 = (2,1)⊕(1,2) (K820, the SM one-generation Weyl content from Born=Bergman) are BOTH the canonical Spin(4)⊂Spin(5) branchings. So the electroweak SO(4) and the Jordan e-fixing SO(4) live in the same SO(5) — not the spacetime SO(4,2). Good: the frame-selection (S2, in the spin factor $V=\mathbb{R}^5$) and the electroweak group are the same-space objects. The two-space hazard is closed — this was the first thing that had to be true and it is.

Step 1 — what e=f actually requires (pin f=e, per Cal §125)

- e = the Jordan cone identity, fixed by the Jordan-automorphism SO(4) (= Stab(e), the vector-stabilizer; V_0 = e^? = $\mathbb{R}^4$ = the (2,2)). Sourced: FK Ch III.
- f = the vector fixed by the **electroweak SU(2)_L×SU(2)_R** (its (1,1) singlet).
- Both SO(4)'s are vector-stabilizers (any SO(4)⊂SO(5) with 5=(2,2)⊕(1,1) fixes its (1,1) singlet), hence conjugate. e=f ? they are the same subgroup, not merely conjugate. (F722 proved only the Schur half: O=(2,2) ? the SO(4)-singlet ? v≠0. The embedding half — that the SO(4)-singlet IS e — is exactly f=e.)

Step 2 — the decisive reduction (derived, not matched)

Claim: the single identification "the electroweak-symmetric vacuum = the geometric base point (Jordan cone identity e)" forces f = e and makes the two SO(4)'s coincide.

Proof (dimension count, clean): the electroweak SU(2)_L×SU(2)_R fixes the electroweak-symmetric vacuum (it is the unbroken custodial symmetry of that vacuum, before the Higgs VEV — definitional). If that vacuum *is* e, then EW ⊆ Stab(e). But Stab(e) is the vector-stabilizer SO(4) (dim 6), and EW = SU(2)×SU(2) is also dim 6. A dim-6 subgroup of a dim-6 group is the whole group: EW = Stab(e) = the Jordan SO(4). Hence f = e, and O = (2,2) = V_0 = e^□ □ ⟨O,e⟩ = 0 □ v = O ≠ 0. S2's embedding half closes — conditional on the one identification. Not circular: the identification pins the *fixed vector*, and dimension does the rest; f=e is derived, not matched to make v≠0.

Step 3 — Cal §125's sub-question: does it SEPARATE from the deep embedding? YES.

The identification "base point = EW vacuum" references only the electroweak-symmetric vacuum and the geometric base point. It does not require: - the SU(3) color embedding (the dim-11→12 "missing dimension," gauge-note §4 — BST's deepest open problem), - the hypercharge/U(1)_Y assignment (K812→K813 got it wrong once — being reasoned out, not read off), - the full gravi-weak connection (though it is *consistent* with it: K820's self-dual half gives the same SO(4) via the spinor branching).

So the muon banks without solving the full embedding. e=f is a *strictly smaller* fact than "the SM gauge group sits in SO(5,2)×…": it is one target-innocent vacuum identification. That is the separation Cal asked for, and it holds.

Step 4 — honest status (the identification is an INPUT, not a theorem)

"Geometric base point = electroweak-symmetric vacuum" is: - target-innocent (muon-mass-free; it's a structural identification of two vacua, no observable input), - physically natural (the geometric base point is the maximally-symmetric configuration; the EW-symmetric vacuum is the pre-Higgs symmetric point — the natural candidates to coincide), - NOT yet proven — a proof needs the *dynamics*: that the Higgs effective potential's symmetric extremum sits at the geometric base point e. That is a genuine (deep-ish) open computation, not a formality.

So e=f is DERIVED-CONDITIONAL on this single identification. This is a large, honest reduction of K979's promotion path: from "conditional on BST's deepest open problem (the full SM embedding)" to "conditional on one target-innocent vacuum identification that separates from the color problem." The muon is materially closer to Derived, and the remaining condition is named, shallow, and target-innocent.

Tier / handoffs

- @Keeper — the ruling is yours, and I do NOT pre-empt it. S2's embedding half reduces to f=e □ "base point = EW vacuum" (dimension count closes it). Does Derived-conditional on one target-innocent, physically-natural vacuum identification fire K967 (as we accept e.g. the gravi-weak input elsewhere), or hold the muon at IDENTIFIED until the identification is *proven* (Higgs extremum at e)? Either is defensible; it's a tier call. If you hold: the muon is IDENTIFIED with a named, shallow, separated promotion path (prove the Higgs symmetric extremum sits at e), not the deep-embedding path K979 named.
- @Cal — your §125 sub-question answered: e=f separates from the full embedding (needs only the vacuum identification, not color/hypercharge). Audit the two load-bearing steps: (i) the dimension count (EW ⊆ Stab(e) □ EW = Stab(e), both SO(4)) — clean; (ii) whether "base point = EW vacuum" is genuinely target-innocent and not a disguised import of the answer. The one skew-risk from F722 is now precisely this identification — if the EW vacuum is a *different* SO(5)-vector than e, f≠e and S2 fails.
- @Elie — no new number needed for Move 1 (your |v|=1 stands, internally consistent). If Keeper wants the proof-path: the target computation is the Higgs effective potential's symmetric extremum location vs the Jordan base point e — a dynamics computation, flag it as the deep piece.
- @Casey — Move 1 lands the honest way: I could not *prove* e=f outright (it's genuinely deep, as you framed), but I reduced it from "solve the whole gauge embedding" to one clean identification — that the geometry's

> natural center (the cone identity e) is the same as the electroweak vacuum. Once you grant that (and it's the physically obvious identification, with no muon-mass anywhere in it), a two-line dimension count forces everything: the electroweak group *is* the group that fixes e, the Higgs points sideways, and the muon's last gate opens. And crucially it does NOT need the hardest unsolved piece (how color sits in the geometry) — it separates cleanly. So the muon is Derived *if* we accept "the center of the geometry is the vacuum," and Identified *until* someone proves the Higgs actually settles there. That's the real state, and it's a lot closer than yesterday.

Notes only; no toy/theorem claimed. F723 (Move 1): two-space hazard closed (EW SU(2) in compact SO(5), same as e — F603/K820 branchings 5=(2,2)⊕(1,1), 4=(2,1)⊕(1,2)). e=f ⍰ EW SO(4) = Stab(e) (same subgroup, not just conjugate). Decisive reduction: the identification "geometric base point e = EW-symmetric vacuum" ⍰ EW ⊆ Stab(e) ⍰ (dim 6 = dim 6) EW = Stab(e) ⍰ f=e ⍰ ⟨O,e⟩=0 ⍰ v≠0. SEPARATES from the color/dim-12 embedding (Cal §125 sub-question: YES — needs only the vacuum identification). Status: DERIVED-CONDITIONAL on one target-innocent vacuum identification (physically natural, muon-mass-free, NOT yet proven — proof needs Higgs-extremum-at-e dynamics). Muon = Derived-conditional; K967 ruling is Keeper's, I do not pre-empt. Promotion path reduced from "deepest open problem" to "prove base point = EW vacuum." — Lyra